

SHAIKH ZEESHAN MURSHED

Senior Frontend Engineer | React | TypeScript | Full-Stack

Montréal, QC, Canada | Open to Remote | sk.zeeshan1992@gmail.com | linkedin.com/in/skizsk | github.com/sk-izsk | izsk.dev | izsk.hashnode.dev

Languages: English, Bengali, Hindi

PROFESSIONAL SUMMARY

Senior Frontend Engineer with 7+ years building high-performance, accessible web interfaces for enterprise SaaS platforms. Deep expertise in React, TypeScript, and component architecture from designing reusable UI libraries consumed across multiple teams to translating complex UX/UI mockups into maintainable, production-grade interfaces. Consistent track record of optimizing front-end performance, instrumenting analytics across user journeys, and integrating cleanly with REST APIs and backend teams. Recently extended into full-stack and AI engineering: built and deployed a FastAPI backend powering a portfolio Ask AI feature, with LLM inference, Upstash Redis caching, IP-based rate limiting, and a structured prompt pipeline. Comfortable owning technical decisions end-to-end, contributing to agile ceremonies, and raising the quality bar through code review, documentation, and mentoring.

TECHNICAL SKILLS

Languages: JavaScript (ES2022+), TypeScript, Python, HTML5, CSS3

Frameworks: React, Next.js, Node.js, Express, FastAPI, GraphQL, React Native (Expo)

State / Data: TanStack Query (React Query), Redux Toolkit, Zustand

UI Libraries: Material-UI (MUI), Ant Design, Tailwind CSS, Shadcn, NativeWind, Styled-Components, Bootstrap, JSS

Testing: Jest, Vitest, Cypress (E2E), React Testing Library, React Native Testing Library, Enzyme, Mocha, Chai, Nock

Tools / DevOps: Git, GitLab CI/CD, Docker, Webpack, Vite, Storybook, Postman, Figma, Jira, Notion, Bun

Visualization / 3D: Three.js, React Three Fiber, Drei, Maf, Framor Motion

Real-Time: Socket.IO, WebSockets, Server-Sent Events (SSE)

Analytics / Monitoring: Frontend event tracking, user funnel instrumentation, error monitoring (Sentry), performance metrics (Core Web Vitals)

Databases: PostgreSQL, MongoDB, Drizzle ORM, Upstash Redis

Auth / Cloud: Clerk, JWT, AWS (S3, Lambda), Cloudinary, Heroku, Netlify, Render

AI & Agentic Tooling: Prompt engineering, LLM integration (Groq, OpenAI-compatible APIs), FastAPI LLM backends, Upstash Redis caching, IP rate limiting, MCP (custom servers, Storybook, Figma, GitHub), agentic workflow automation, AI code review

EXPERIENCE

Independent Projects & AI Engineering

Mar 2026 – Present

Self-directed

Full-stack and AI engineering development during career transition

- Built an end-to-end Ask AI portfolio feature: FastAPI backend integrating LLM inference APIs, Upstash Redis caching with versioned cache keys, IP-based rate limiting (3 req/IP/hour), and a structured prompt pipeline enforcing resume-grounded answers with JSON-only output.
- Shipped a React 19 / TypeScript / Vite chat frontend with cold-start handling, typing indicators, source dropdowns, auto-growing input, keyboard shortcuts, and backend error surfacing.
- Deepening AI engineering foundations: RAG pipelines, vector databases, LLM orchestration patterns, and agentic workflows – applied through self-directed projects and technical writing.
- Deepening backend engineering skills through deliberate practice: built real-time backends with Express 5, Socket.IO, and MongoDB; REST APIs with FastAPI, PostgreSQL, and Drizzle ORM – covering auth patterns, caching, deployment, and API design end to end.
- Participated in AI evaluation and survey projects for external organizations, staying active in the AI engineering space during the transition.
- Maintained a technical blog at izsk.hashnode.dev covering advanced React patterns and AI engineering topics.

Software Engineer

Nov 2020 – Mar 2026

SwiftConnect — Montréal, QC (position eliminated, business restructuring)

Front-end lead on enterprise access & security management SaaS platform

- Designed and evolved high-performance web interfaces using React, TypeScript, Styled-Components, Ant Design, Redux Toolkit, and TanStack Query, covering complex authorization workflows across multiple product areas.
- Translated UX/UI mockups into reusable, accessible component systems (WCAG standards) across 10+ UI components, cutting design-to-dev revision cycles by ~30%.
- Authored and maintained a private, Storybook-documented NPM component library adopted across cross-functional teams, reducing duplicate front-end code by ~40% and enforcing consistent UI standards at scale.

- Architected and published a private Node.js SDK (core JavaScript, internal NPM registry) wrapping all backend API calls behind a versioned, typed interface eliminating duplicated HTTP logic across projects and distributed to third-party customers integrating SwiftConnect's API into their own systems.
- Built a companion React SDK layered over the Node SDK, exposing React-specific hooks and components for API integration adopted by external customers building custom frontends against SwiftConnect's access management API.
- Solely designed, built, and maintained an internal React Native (Expo) dashboard application for the customer-facing team as the only engineer on the team with prior React Native experience set up the project from scratch, including navigation, API integration, data-heavy dashboard views, and full unit test coverage via React Native Testing Library.
- Engineered a React-based email templating pipeline where developers author SendGrid templates as React components; merged changes trigger automated CI/CD deployment to SendGrid, keeping all email output version-controlled and eliminating manual template management.
- Instrumented front-end analytics and KPI tracking across key user journeys, giving product teams real-time visibility into feature adoption, funnel drop-off, and performance metrics.
- Optimized front-end performance and code quality through GitLab CI/CD pipeline setup, Core Web Vitals tracking, and consistent test coverage via Jest and React Testing Library.
- Instrumented Sentry error monitoring across all front-end surfaces, configuring release tracking, source map uploads, and alert routing to reduce mean time to detection on production issues.
- Mentored 2-3 junior developers through code reviews, pair programming, and bi-weekly knowledge-sharing sessions, reducing review turnaround time by ~25%.
- Acted as technical lead and proxy Product Owner during critical delivery phases, translating business requirements into architecture plans and coordinating sprint execution across front-end and backend teams.
- Built a Storybook + Figma MCP pipeline for commit-time design-integrity validation, and engineered custom MCP servers enabling in-editor API doc queries, reducing cross-functional revision cycles by ~35%.
- Integrated AI tooling into engineering workflows: LLM-assisted PR reviews via GitHub MCP, agentic scaffolding for boilerplate and test generation, all with human-review gates.

Full Stack Developer

Oct 2020 – Nov 2020

Hatchways — Remote

Agile team of 4, full-stack web application development

- Built web applications using React and Material-UI on the front end and Express, Node.js, MongoDB, Cloudinary, and JWT authentication on the backend.
- Delivered full-stack features in 2-week sprint cycles within a cross-functional Agile team of 4.

Junior Software Developer

July 2019 – June 2020

Guiker Inc. — Montréal, QC

Mobile-first apartment rental marketplace startup

- Contributed to a full React rewrite of the main web product using TypeScript, Material-UI, RESTful APIs, and JSS with comprehensive Jest test coverage.
- Integrated REST API endpoints and owned specific Node.js/Express routing and middleware logic, directly reducing front-end-to-backend integration issues.
- Implemented in-app landlord-tenant chat and lease-signing flows, contributing to a 15–20% increase in repeat session activity.
- Developed mobile application features for iOS and Android using React Native (Expo), TypeScript, and Jest unit testing.

PROJECTS

SwiftConnect | Enterprise Access & Security Management Platform

- **Core Web Client:** Designed the primary access control interface in React/TypeScript. Built global state and caching layers (Redux Toolkit, TanStack Query) to manage complex authorization data, keep the UI stable during heavy mutations, and surface real-time analytics on feature adoption and conversion funnels.
- **Internal React Native Dashboard:** Solo-built an internal React Native (Expo) application for the customer team from scratch project setup, routing, API integration layered on the existing Node/React SDK and design system, and data-heavy dashboard screens, all covered by unit tests with React Native Testing Library.
- **SDK Distribution:** Architected a Node.js SDK and companion React SDK (private NPM) exposing typed, versioned API interfaces adopted internally across all projects and distributed externally to third-party customers building custom frontends against SwiftConnect's access management API.
- **Design System:** Authored a private, strongly-typed NPM component library enforced across cross-functional teams, reducing duplicate front-end code by ~40% and maintaining consistent UI standards across the platform.
- **Front-end Quality & Observability:** Established CI/CD pipelines, Sentry error monitoring (release tracking, source maps, alert routing), and Core Web Vitals performance tracking across all front-end surfaces.

- **AI-Powered Developer Experience:** Built custom MCP servers (Storybook, Figma, GitHub) to automate design validation, streamline PR workflows, and eliminate developer context-switching.

Portfolio Ask AI | *Full-Stack LLM Feature – React Frontend + FastAPI Backend* | izsk.dev | github.com/sk-izsk/zsk-ai-resume-api

- Built an end-to-end Ask AI feature: React 19 / TypeScript / Vite frontend with a /ask-ai route, and a Python FastAPI backend on Render integrating Groq’s llama-3.3-70b-versatile model for resume-grounded question answering.
- Designed a prompt pipeline enforcing strict answer grounding with a blocked response field for off-topic queries and JSON-only structured output (answer, sources, blocked).
- Implemented Upstash Redis REST-based caching with versioned cache keys for prompt/behavior invalidation, and IP-based rate limiting (3 requests/IP/hour) with 429 error handling propagated to the frontend.
- Built a compact_profile() pipeline sending reduced resume context to Groq instead of the full 54KB JSON, cutting token usage and keeping inference costs manageable.
- Frontend handles Render free-tier cold starts with a /health pre-flight call blocking the chat panel with a loading overlay until the backend wakes, then fires POST /chat.

Whisper Chat | *Full-Stack Real-Time Messaging App* | [Live Demo](https://live-demo.github.com/sk-izsk/react-native-fullstack-whisper-chat) | github.com/sk-izsk/react-native-fullstack-whisper-chat

- Built a real-time chat system across three surfaces: a React 19 / Vite web client, a React Native / Expo / Sentry mobile client, and a shared Express 5 / Socket.IO / MongoDB TypeScript backend, with Clerk authentication on all three.
- Split state across two layers: TanStack Query for server state and Zustand for transient real-time state (socket connection, unread markers, typing indicators, online presence), keeping cache invalidation isolated from socket event handling.
- Refactored socket logic into dedicated modules and split oversized screens into focused hooks and components; added optimistic message sending, Sentry monitoring on mobile, and consistent behavior across web and native without duplicating business logic.

Z-Graph | *Interactive Math Visualization Platform* | z-graph.vercel.app | github.com/sk-izsk/z-graph

- Built a multi-mode graphing platform in React 18 / TypeScript covering 17 mathematical categories algebra, calculus, polar, parametric, vector fields, Fourier series, Julia/Mandelbrot fractals, and 3D implicit surfaces with live formula editing, parameter sliders, animation controls, and a preset navigation system.
- Combined 2D rendering via Mafs and 3D via Three.js / React Three Fiber under a shared Zustand state layer; mathjs handles live formula parsing; Web Workers offload fractal sampling and surface meshing off the main thread.

ZTable Web | *Interactive Chemistry App* | zsk-periodic.vercel.app | github.com/sk-izsk/zsk-periodic

- Designed a performance-focused render pipeline: equation balancing runs off the main thread in a Web Worker, route screens lazy-load through TanStack Router, and 118 periodic table cells use memoized Zustand selectors to prevent broad re-renders on interaction.
- Built with React 19, TypeScript, Vite, and Bun across 8 features including 3D atom visualization (Three.js / React Three Fiber), molar mass parsing, solubility lookup, PDF worksheet export, and i18n.

Productify | *Full-Stack Product Management App* | izsk.dev

- Built a full-stack React 19 and Node.js application using Vite, React Router 7, and compound component pattern for a scalable, maintainable UI.
- Architected a strongly-typed REST API using Express 5, Drizzle ORM, and PostgreSQL (Neon) with TanStack Query for cache-aware mutations and Clerk for end-to-end authentication and role-based authorization.

EDUCATION

Concordia University

M.Eng., Electrical & Computer Engineering

Montréal, QC

Sept 2016 – Dec 2018

- Relevant coursework: Computer Architecture Design, Microprocessor Applications, Digital Signal Processing, Model-Driven Software Engineering, Formal Hardware Verification, Real-Time Applications, Robotics.

DecodeMTL

Full-Stack Web Development Bootcamp

Montréal, QC

April 2019 – May 2019

- Intensive full-stack web development program covering JavaScript, React, Node.js, Express, MongoDB, HTML5, CSS3.